Notebook Export

Cell 1 (code)

```
import json
import pandas as pd

def jsons_to_df(json_paths, method_names, columns=None):
    rows = []

    for path, name in zip(json_paths, method_names):
        try:
            with open(path, "r") as f:
                d = json.load(f)
        except FileNotFoundError:
            print(f"File not found {name}: {path}")
            d = {col: None for col in columns}
        d["method"] = name
        rows.append(d)

    df = pd.DataFrame(rows)

    if columns:
        df = df[["method"] + columns]

    display(df)
    # return df

# columns = ['roc_auc_mean', 'roc_auc_std', 'accuracy_mean', 'accuracy_std']
columns = ['accuracy_mean', 'accuracy_std', 'precision_mean', 'precision_std', 'recall_mean', 'recal
columns_m_specificity = ['accuracy_mean', 'accuracy_std', 'precision_mean', 'precision_std', 'recall_
method_names = ['Radiomics Attention Pooling', 'Radiomics Statistical Pooling', '2D images Googlenet
```

Cell 2 (code)

```
def root_path_changer(json_paths, old_root, new_root):
    new_paths = []
    for path in json_paths:
        new_path = path.replace(old_root, new_root)
        new_paths.append(new_path)
    return new_paths

new_root = "../"
old_root = "/mnt/l/Basic/divi/jstoker/slicer_pdac/Marius/Reza Morphology/codes/"
```

Cell 3 (code)

```
json_paths = [
    "/mnt/l/Basic/divi/jstoker/slicer_pdac/Marius/Reza Morphology/codes/Results/RadiomicsMIL/overall_
    "/mnt/l/Basic/divi/jstoker/slicer_pdac/Marius/Reza Morphology/codes/Results/StatisticalPoolingML
    "/mnt/l/Basic/divi/jstoker/slicer_pdac/Marius/Reza Morphology/codes/Results/GooglenetLSTM/overal
    "/mnt/l/Basic/divi/jstoker/slicer_pdac/Marius/Reza Morphology/codes/Results/RPNet3D/overall_survi
    "",
    ]
json_paths = root_path_changer(json_paths, old_root, new_root)
print("overall survival results:")
jsons_to_df(json_paths, method_names, columns)
Output:
overall survival results:
Output:
                          method  accuracy_mean  accuracy_std  precision_mean  \
0    Radiomics Attention Pooling       0.710594      0.037687        0.705960
1  Radiomics Statistical Pooling       0.625397      0.050261        0.678925
2            2D images Googlenet       0.628172      0.045039        0.660734
3                        RPNet3D       0.678624      0.048136        0.676964

   precision_std  recall_mean  recall_std  specificity_mean  specificity_std  \
0       0.058908     0.877303    0.040880          0.474571         0.054134
1       0.113199     0.784130    0.175592          0.416886         0.313499
2       0.058659     0.778971    0.132387          0.416654         0.165276
3       0.077915     0.878340    0.064022          0.405234         0.113823

   f1_score_mean  f1_score_std  roc_auc_mean  roc_auc_std
0       0.780641      0.036731      0.741556     0.034693
```

```
1       0.707031      0.045592      0.673124     0.081794
2       0.708472      0.048454      0.656022     0.062588
3       0.761137      0.044405      0.679033     0.078739
```

Cell 4 (code)

```
json_paths = [
    "/mnt/l/Basic/divi/jstoker/slicer_pdac/Marius/Reza Morphology/codes/Results/RadiomicsMIL/early_re
    "/mnt/l/Basic/divi/jstoker/slicer_pdac/Marius/Reza Morphology/codes/Results/StatisticalPoolingMLL
    "/mnt/l/Basic/divi/jstoker/slicer_pdac/Marius/Reza Morphology/codes/Results/GooglenetLSTM/early_r
     "/mnt/l/Basic/divi/jstoker/slicer_pdac/Marius/Reza Morphology/codes/Results/RPNet3D/early_recurr
    "",
    ]
json_paths = root_path_changer(json_paths, old_root, new_root)
print("early recurrence results:")
jsons_to_df(json_paths, method_names, columns)
Output:
early recurrence results:
Output:
                          method  accuracy_mean  accuracy_std  precision_mean  \
0    Radiomics Attention Pooling       0.528245      0.055108        0.384222
1  Radiomics Statistical Pooling       0.627485      0.077048        0.462119
2            2D images Googlenet       0.555892      0.089715        0.352655
3                        RPNet3D       0.561839      0.129789        0.313487

   precision_std  recall_mean  recall_std  specificity_mean  specificity_std  \
0       0.075958     0.771514    0.086373          0.425651         0.081773
1       0.099635     0.696945    0.138288          0.605319         0.167179
2       0.036214     0.516138    0.305882          0.546563         0.318476
3       0.195330     0.547896    0.337396          0.554321         0.306069

   f1_score_mean  f1_score_std  roc_auc_mean  roc_auc_std
0       0.505607      0.053656      0.652296     0.046876
1       0.538720      0.042642      0.703593     0.070600
2       0.397769      0.118333      0.570834     0.044540
3       0.384601      0.219168      0.560333     0.042152
```

Cell 5 (code)

```
json_paths = [
    "/mnt/l/Basic/divi/jstoker/slicer_pdac/Marius/Reza Morphology/codes/Results/RadiomicsMIL/morph_re
    "/mnt/l/Basic/divi/jstoker/slicer_pdac/Marius/Reza Morphology/codes/Results/StatisticalPoolingMLL
    "/mnt/l/Basic/divi/jstoker/slicer_pdac/Marius/Reza Morphology/codes/Results/GooglenetLSTM/morph_r
        "/mnt/l/Basic/divi/jstoker/slicer_pdac/Marius/Reza Morphology/codes/Results/RPNet3D/morph_res
    "",
    ]
json_paths = root_path_changer(json_paths, old_root, new_root)
print("morphology response results (3 classes):")
jsons_to_df(json_paths, method_names, columns_m_specificity)
Output:
morphology response results (3 classes):
Output:
                          method  accuracy_mean  accuracy_std  precision_mean  \
0    Radiomics Attention Pooling       0.383518      0.093554        0.397340
1  Radiomics Statistical Pooling       0.409946      0.085947        0.500704
2            2D images Googlenet       0.435928      0.137905        0.539683
3                        RPNet3D       0.481759      0.058609        0.425341

   precision_std  recall_mean  recall_std  f1_score_mean  f1_score_std  \
0       0.181688     0.383518    0.093554       0.361084      0.144181
1       0.031941     0.409946    0.085947       0.411795      0.097364
2       0.118570     0.435928    0.137905       0.409884      0.149738
3       0.112603     0.481759    0.058609       0.445777      0.089476

   roc_auc_mean  roc_auc_std
0      0.554182     0.044965
1      0.546242     0.023595
2      0.549146     0.051689
3      0.524799     0.048939
```

Cell 6 (code)

```
json_paths = [
    "/mnt/l/Basic/divi/jstoker/slicer_pdac/Marius/Reza Morphology/codes/Results/RadiomicsMIL/patholog
    "/mnt/l/Basic/divi/jstoker/slicer_pdac/Marius/Reza Morphology/codes/Results/StatisticalPoolingMLL
    "/mnt/l/Basic/divi/jstoker/slicer_pdac/Marius/Reza Morphology/codes/Results/GooglenetLSTM/patholo
```

```
        "/mnt/l/Basic/divi/jstoker/slicer_pdac/Marius/Reza Morphology/codes/Results/RPNet3D/patholog
    "",
    ]
json_paths = root_path_changer(json_paths, old_root, new_root)
print("pathology results:")
jsons_to_df(json_paths, method_names, columns)
Output:
pathology results:
Output:
                          method  accuracy_mean  accuracy_std  precision_mean  \
0    Radiomics Attention Pooling       0.587209      0.079473        0.602034
1  Radiomics Statistical Pooling       0.581959      0.033956        0.605883
2            2D images Googlenet       0.549057      0.065545        0.561877
3                        RPNet3D       0.530470      0.081000        0.549053

   precision_std  recall_mean  recall_std  specificity_mean  specificity_std  \
0       0.084664     0.915561    0.098212          0.152036         0.233429
1       0.034871     0.767114    0.188489          0.300116         0.250058
2       0.053632     0.910606    0.157496          0.052632         0.117688
3       0.075255     0.769038    0.258451          0.184390         0.215212

   f1_score_mean  f1_score_std  roc_auc_mean  roc_auc_std
0       0.719093      0.045543      0.635184     0.112620
1       0.669851      0.084711      0.614471     0.055163
2       0.692378      0.082212      0.585261     0.073997
3       0.632045      0.151279      0.469310     0.140071
```

Cell 7 (code)

```
json_paths = [
    "/mnt/l/Basic/divi/jstoker/slicer_pdac/Marius/Reza Morphology/codes/Results/MorphScoreRadiomicsM
    "/mnt/l/Basic/divi/jstoker/slicer_pdac/Marius/Reza Morphology/codes/Results/MorphScoreStatistica
    "/mnt/l/Basic/divi/jstoker/slicer_pdac/Marius/Reza Morphology/codes/Results/MorphScoreGooglenetL
    "/mnt/l/Basic/divi/jstoker/slicer_pdac/Marius/Reza Morphology/codes/Results/MorphScoreRPNet3D/fi
    "",
    ]
json_paths = root_path_changer(json_paths, old_root, new_root)
print(" morphology score results (3 classes):")
method_names = ['MorphScore Attention Pooling (all features)', 'MorphScore Statistical Pooling (all 
jsons_to_df(json_paths, method_names, columns_m_specificity)
Output:
 morphology score results (3 classes):
Output:
                                          method  accuracy_mean  accuracy_std  \
0    MorphScore Attention Pooling (all features)       0.348021      0.085555
1  MorphScore Statistical Pooling (all features)       0.387961      0.127181
2                          MorphScore GooglenetLSTM       0.438374      0.046953
3                                        RPNet3D       0.444923      0.086884

   precision_mean  precision_std  recall_mean  recall_std  f1_score_mean  \
0        0.415689       0.048606     0.348021    0.085555       0.343538
1        0.489755       0.057919     0.387961    0.127181       0.346217
2        0.405299       0.087344     0.438374    0.046953       0.391627
3        0.463931       0.019201     0.444923    0.086884       0.398881

   f1_score_std  roc_auc_mean  roc_auc_std
0      0.086483      0.551342     0.022462
1      0.128318      0.561012     0.043552
2      0.026989      0.558728     0.055488
3      0.121916      0.585052     0.060719
```

Cell 8 (markdown)

morph score bevacizumab

Cell 9 (code)

```
# columns_msb = ['roc_auc', 'accuracy']
columns_msb = ['accuracy', 'precision', 'recall', "f1_score", "roc_auc"]
method_names = ['Radiomics Attention Pooling', 'Radiomics Statistical Pooling', '2D images Googlenet
json_paths = [
    "/mnt/l/Basic/divi/jstoker/slicer_pdac/Marius/Reza Morphology/codes/Results/MorphScoreRadiomicsM
    "/mnt/l/Basic/divi/jstoker/slicer_pdac/Marius/Reza Morphology/codes/Results/MorphScoreStatistica
    "/mnt/l/Basic/divi/jstoker/slicer_pdac/Marius/Reza Morphology/codes/Results/MorphScoreGooglenetL
    "/mnt/l/Basic/divi/jstoker/slicer_pdac/Marius/Reza Morphology/codes/Results/MorphScoreRPNet3D/be
```

```
    "/mnt/l/Basic/divi/jstoker/slicer_pdac/Marius/Reza Morphology/codes/Results/MorphScore3DMTLAtten
    ]
json_paths = root_path_changer(json_paths, old_root, new_root)
print("Bevacizumab morphology score results (3 classes):")
method_names = ['MorphScore Attention Pooling (all features)', 'MorphScore Statistical Pooling (all f

jsons_to_df(json_paths, method_names, columns_msb)
Output:
Bevacizumab morphology score results (3 classes):
File not found 3D MTL Attention Siamese: ../Results/MorphScore3DMTLAttentionSiamese/bevacizumab_morph
Output:
                                          method  accuracy  precision  \
0    MorphScore Attention Pooling (all features)  0.348910   0.417880
1  MorphScore Statistical Pooling (all features)  0.390966   0.456064
2                        MorphScore GooglenetLSTM  0.435583   0.415668
3                                        RPNet3D  0.435185   0.457076
4                        3D MTL Attention Siamese       NaN        NaN

     recall  f1_score   roc_auc
0  0.348910  0.365646  0.551476
1  0.390966  0.409490  0.558900
2  0.435583  0.423226  0.559328
3  0.435185  0.443460  0.584068
4       NaN       NaN       NaN
```

Cell 10 (markdown)

Radiomics feature set analysis

Cell 11 (code)

```
json_paths = [
    "/mnt/l/Basic/divi/jstoker/slicer_pdac/Marius/Reza Morphology/codes/Results/MorphScoreStatistica
    "/mnt/l/Basic/divi/jstoker/slicer_pdac/Marius/Reza Morphology/codes/Results/MorphScoreStatistica
    "/mnt/l/Basic/divi/jstoker/slicer_pdac/Marius/Reza Morphology/codes/Results/MorphScoreStatistica
    "/mnt/l/Basic/divi/jstoker/slicer_pdac/Marius/Reza Morphology/codes/Results/MorphScoreStatistica
    "/mnt/l/Basic/divi/jstoker/slicer_pdac/Marius/Reza Morphology/codes/Results/MorphScoreStatistica
    ]
json_paths = root_path_changer(json_paths, old_root, new_root)
print("morphology score results (3 classes):")
method_names = ['StatisticalPoolingMLP (all features)',
                'StatisticalPoolingMLP (boundary_intensity_texture)',
                'StatisticalPoolingMLP (shape_intensity_texture)',
                'StatisticalPoolingMLP (shape_boundary_texture)',
                'StatisticalPoolingMLP (shape_boundary_intensity)']
jsons_to_df(json_paths, method_names, columns_m_specificity)
Output:
morphology score results (3 classes):
Output:
                                              method  accuracy_mean  \
0               StatisticalPoolingMLP (all features)       0.387961
1  StatisticalPoolingMLP (boundary_intensity_text...       0.379981
2    StatisticalPoolingMLP (shape_intensity_texture)       0.345622
3     StatisticalPoolingMLP (shape_boundary_texture)       0.390711
4   StatisticalPoolingMLP (shape_boundary_intensity)       0.389646

   accuracy_std  precision_mean  precision_std  recall_mean  recall_std  \
0      0.127181        0.489755       0.057919     0.387961    0.127181
1      0.092186        0.408269       0.167169     0.379981    0.092186
2      0.121751        0.359196       0.193692     0.345622    0.121751
3      0.105230        0.492825       0.064051     0.390711    0.105230
4      0.058068        0.482533       0.038299     0.389646    0.058068

   f1_score_mean  f1_score_std  roc_auc_mean  roc_auc_std
0       0.346217      0.128318      0.561012     0.043552
1       0.344048      0.140673      0.568466     0.024756
2       0.323887      0.165037      0.559785     0.035537
3       0.388945      0.107622      0.577589     0.055358
4       0.405257      0.053064      0.591919     0.029522
```

Cell 12 (code)

```
json_paths = [
    "/mnt/l/Basic/divi/jstoker/slicer_pdac/Marius/Reza Morphology/codes/Results/MorphScoreStatistica
    "/mnt/l/Basic/divi/jstoker/slicer_pdac/Marius/Reza Morphology/codes/Results/MorphScoreStatistica
```

```
    "/mnt/l/Basic/divi/jstoker/slicer_pdac/Marius/Reza Morphology/codes/Results/MorphScoreStatistica
    "/mnt/l/Basic/divi/jstoker/slicer_pdac/Marius/Reza Morphology/codes/Results/MorphScoreStatistica
    ]
json_paths = root_path_changer(json_paths, old_root, new_root)
print("morphology score results (3 classes):")
method_names = [
                'StatisticalPoolingMLP (shape)',
                'StatisticalPoolingMLP (boundary)',
                'StatisticalPoolingMLP (intensity)',
                'StatisticalPoolingMLP (texture)']
jsons_to_df(json_paths, method_names, columns_m_specificity)
Output:
morphology score results (3 classes):
Output:
                              method  accuracy_mean  accuracy_std  \
0      StatisticalPoolingMLP (shape)       0.450952      0.041049
1   StatisticalPoolingMLP (boundary)       0.428684      0.044922
2  StatisticalPoolingMLP (intensity)       0.418784      0.095452
3    StatisticalPoolingMLP (texture)       0.324585      0.097208

   precision_mean  precision_std  recall_mean  recall_std  f1_score_mean  \
0        0.475661       0.056092     0.450952    0.041049       0.453584
1        0.464506       0.022809     0.428684    0.044922       0.428137
2        0.469669       0.066841     0.418784    0.095452       0.417493
3        0.407167       0.064101     0.324585    0.097208       0.297615

   f1_score_std  roc_auc_mean  roc_auc_std
0      0.041836      0.582806     0.043387
1      0.038813      0.577271     0.027504
2      0.090386      0.576186     0.071324
3      0.090295      0.528454     0.025809
```

Cell 13 (markdown)

MIL

Cell 14 (code)

```
json_paths = [
    "/mnt/l/Basic/divi/jstoker/slicer_pdac/Marius/Reza Morphology/codes/Results/MorphScoreRadiomicsM
    "/mnt/l/Basic/divi/jstoker/slicer_pdac/Marius/Reza Morphology/codes/Results/MorphScoreRadiomicsM
    "/mnt/l/Basic/divi/jstoker/slicer_pdac/Marius/Reza Morphology/codes/Results/MorphScoreRadiomicsM
    "/mnt/l/Basic/divi/jstoker/slicer_pdac/Marius/Reza Morphology/codes/Results/MorphScoreRadiomicsM
    "/mnt/l/Basic/divi/jstoker/slicer_pdac/Marius/Reza Morphology/codes/Results/MorphScoreRadiomicsM
    ]
json_paths = root_path_changer(json_paths, old_root, new_root)
print("morphology score results (3 classes):")
method_names = ['MIL (all features)',
                'MIL (boundary_intensity_texture)',
                'MIL (shape_intensity_texture)',
                'MIL (shape_boundary_texture)',
                'MIL (shape_boundary_intensity)']
jsons_to_df(json_paths, method_names, columns_m_specificity)
Output:
morphology score results (3 classes):
Output:
                             method  accuracy_mean  accuracy_std  \
0                MIL (all features)       0.348021      0.085555
1  MIL (boundary_intensity_texture)       0.389424      0.064114
2     MIL (shape_intensity_texture)       0.452082      0.040612
3      MIL (shape_boundary_texture)       0.379498      0.110336
4    MIL (shape_boundary_intensity)       0.408415      0.107566

   precision_mean  precision_std  recall_mean  recall_std  f1_score_mean  \
0        0.415689       0.048606     0.348021    0.085555       0.343538
1        0.437074       0.050688     0.389424    0.064114       0.374428
2        0.453179       0.051680     0.452082    0.040612       0.437180
3        0.370409       0.147387     0.379498    0.110336       0.327708
4        0.450552       0.066239     0.408415    0.107566       0.400380

   f1_score_std  roc_auc_mean  roc_auc_std
0      0.086483      0.551342     0.022462
1      0.063606      0.539950     0.035102
2      0.051123      0.560620     0.042933
3      0.126046      0.540371     0.047286
4      0.090162      0.567779     0.049525
```

Cell 15 (code)

```
json_paths = [
    "/mnt/l/Basic/divi/jstoker/slicer_pdac/Marius/Reza Morphology/codes/Results/MorphScoreRadiomicsMI
    "/mnt/l/Basic/divi/jstoker/slicer_pdac/Marius/Reza Morphology/codes/Results/MorphScoreRadiomicsMI
    "/mnt/l/Basic/divi/jstoker/slicer_pdac/Marius/Reza Morphology/codes/Results/MorphScoreRadiomicsMI
    "/mnt/l/Basic/divi/jstoker/slicer_pdac/Marius/Reza Morphology/codes/Results/MorphScoreRadiomicsMI
    ]
json_paths = root_path_changer(json_paths, old_root, new_root)
print("morphology score results (3 classes):")
method_names = [
                'RadiomicsMIL (shape)',
                'RadiomicsMIL (boundary)',
                'RadiomicsMIL (intensity)',
                'RadiomicsMIL (texture)']
jsons_to_df(json_paths, method_names, columns_m_specificity)
Output:
morphology score results (3 classes):
Output:
                     method  accuracy_mean  accuracy_std  precision_mean  \
0      RadiomicsMIL (shape)       0.396273      0.038260        0.457363
1   RadiomicsMIL (boundary)       0.407729      0.061468        0.395896
2  RadiomicsMIL (intensity)       0.408681      0.087534        0.396787
3    RadiomicsMIL (texture)       0.351402      0.065361        0.436319

   precision_std  recall_mean  recall_std  f1_score_mean  f1_score_std  \
0       0.051799     0.396273    0.038260       0.403903      0.024810
1       0.065799     0.407729    0.061468       0.373177      0.064816
2       0.154892     0.408681    0.087534       0.377990      0.122808
3       0.154055     0.351402    0.065361       0.283673      0.116899

   roc_auc_mean  roc_auc_std
0      0.552853     0.038604
1      0.567396     0.039109
2      0.555989     0.025320
3      0.536297     0.039036
```